BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 086 · Merge Sort

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does insertion sort do with the next unsorted item?

Q2. Which task suits utility software?
A. Writing an essay
B. Computer maintenance
C. Editing a photograph
D. Playing a game

Q3. “File management is only the execution of arithmetic instructions.” Correct this. Explain why.

Q4. A teacher stores private records on a portable drive. “An encryption utility makes a drive weigh less.” Correct this. What does it protect?

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I Do · Read new knowledge

Key vocabulary

Merge sort — Divide-and-conquer sorting method that merges ordered sublists.

Divide and conquer — Split a problem into smaller subproblems, then combine solutions.

Split — Separate the list into smaller parts.

Merge — Combine sorted lists into one ordered list.

Sorted list — Elements arranged in order.

Comparison — Check two values to decide which enters the result next.

New learning

Divide the problem: Merge sort divides a collection into halves until single elements remain.

Single elements are sorted: A list containing one item needs no further sorting.

Merge in order: Merge two sorted halves by comparing their front values, taking the smaller one and repeating.

Rebuild the whole list: The smaller sorted lists combine step by step to produce one completely sorted result. OCR requires the main steps, not code memorisation.

Worked example

Trace merge sort on [8,3,5,1]. Show the split, the sorted pairs and the final sorted list. [4 marks]

1. Split into [8,3] and [5,1] [1].

2. Split into individual elements [8],[3],[5],[1] [1].

3. Merge into sorted pairs [3,8] and [1,5] [1].

4. Merge these pairs in order: [1,3,5,8] [1].

We Do · Practise with guidance

Guided 1

Merge sort [6,2,7,1]. Give the sorted pairs and final result. [4 marks]

Worked method: Split [6,2] and [7,1] [1]. Sort pairs as [2,6] and [1,7] [1]. Compare leading elements when merging [1]. Final order [1,2,6,7] [1].

Guided 2

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Explain why merge sort splits [9,3,4,2] down to single values. [2 marks]

Worked method: Single values are already sorted [1] and can then be merged into correctly ordered larger groups [1].

Guided 3

Merge the sorted lists [2,7] and [3,6] into one sorted result. [4 marks]

Worked method: Compare 2 and 3, take 2 [1]. Compare 7 and 3, take 3 [1]. Compare 7 and 6, take 6 [1]. Then append 7 → [2,3,6,7] [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the condition choosing the smaller front value while merging two sorted lists.

```
array left = [3,8]
array right = [1,5]
if ... then
    print(left[0])
else
    print(right[0])
endif
```

write

Question 1

State the key idea behind the divide step of merge sort. [2 marks] Source: Original OCR J277-style question; not an official past-paper original.

Question 2

Trace merge sort on [4,1,3,2], showing two sorted pairs and the final result. [4 marks] Source: Original OCR J277-style question; not an official past-paper original.

Question 3

Explain the merge-sort process for [7,2,6,1], including split, comparison and merge. [6 marks] Source: Original OCR J277-style question; not an official past-paper original.

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Exit Ticket · Answer before checking

Merge sort combines sorted smaller lists to produce a sorted full list.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Place the next item in the correct position in the sorted part. Shift larger items right.

Q2. B — Computer maintenance

Q3. It manages files and folders: naming, saving, copying, moving and deleting them.

Q4. Encryption uses an algorithm and key to change readable data into ciphertext. It protects privacy; compression reduces size and defragmentation rearranges HDD data.

I Do / We Do pages · Guided practice

Q1. Split [6,2] and [7,1] [1]. Sort pairs as [2,6] and [1,7] [1]. Compare leading elements when merging [1]. Final order [1,2,6,7] [1].

Q2. Single values are already sorted [1] and can then be merged into correctly ordered larger groups [1].

Q3. Compare 2 and 3, take 2 [1]. Compare 7 and 3, take 3 [1]. Compare 7 and 6, take 6 [1]. Then append 7 → [2,3,6,7] [1].

You Do pages · Code First

```
array left = [3,8]
array right = [1,5]
if left[0] < right[0] then
    print(left[0])
else
    print(right[0])
endif
```

Expected result: 1

You Do pages · Independent questions

Q1. Split an array into smaller halves [1] until individual items/single-element lists remain [1].

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Q2. Split into [4,1] and [3,2] [1]; merge singletons into [1,4] [1] and [2,3] [1]; final merge [1,2,3,4] [1].

Q3. Split into [7,2] and [6,1] [1]; split into singletons [1]. Merge into [2,7] [1] and [1,6] [1]. Compare leading values of sorted halves and take smaller repeatedly [1], producing [1,2,6,7] [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: